GANLIN CHEN

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https://scholar.google.com/citations?user=UNWAQRAAAAAJ&hl=en&oi=ao

EDUCATION

University of Michigan, Ann Arbor *Sep. 2021 – Jun. 2026 (expected)*
Doctor of Philosophy in Materials Science & Engineering GPA:**4.0**
Joint degree in Scientific Computing

University of Michigan, Ann Arbor *Sep. 2019 – Apr. 2021*
Master of science in Materials Science & Engineering GPA:**4.0**

Soochow University, China *Sep. 2015 – Jun. 2019*
Bachelor of engineering in New Energy Materials and Devices GPA:**3.88**

RESEARCH EXPERIENCE

Graduate Student Research Assistant *Sep. 2021 –*
University of Michigan, Ann Arbor

- Developed theoretical framework for efficient identifications of competing deformation modes in metastable β Ti alloys.
- Implemented algorithm for structure identifications between ω-β in multicomponet Ti alloys under deformation.
- Conducted temperature-dependent free-energy landscape analysis using metadynamics for β-α" phase transformations.
- Investigated corrosion resistance of intermetallic phases in Al alloys.
- Investigate machine-learning-assisted approaches for efficient estimation of composition- and temperature-dependent misfit strain energy along phase boundaries.(on going)

Research Assistant *Jan. 2020 – Sep. 2021*
University of Michigan, Ann Arbor

- Investigated tunneling effect across oxide layer on current at Metal-Insulator-Electrolyte (MIE) interfaces.
- Investigated effects of defects on choride-induced pitting corrosion of Al_2O_3 and Cr_2O_3 under aqueous environment.

TEACHING EXPERIENCE

- Graduate student instructor of MSE 350, Structural of Materials, fall term of 2024.

PUBLICATIONS

- **Ganlin Chen**, Yang Guo, Liang Qi. Computational studies of cathodic reaction activities on intermetallic particles and solute clusters in aluminum alloys. **Ready to submit to *npj Materials Degradation*.**
- **Ganlin Chen**, Deepak V Pillai, Yufeng Zheng, Liang Qi. Energetic Origins of Competing Deformation Modes in Metastable Titanium Alloys.arXiv preprint arXiv:2510.13113.
- Chao Huang, Eva Katharine Pontrelli, Jae Wan Lee, Binyu Wang, **Ganlin Chen**, Tatiane Cogo Machado, Hemanth Maddali, Benjamin De Chazal, Anish Tuteja, Ronald G Larson, Naír Rodríguez-Hornedo, Max Shtein. ***Advanced Materials***,**47**, e10419 (2025)https://advanced.onlinelibrary.wiley.com/doi/full/10.1002/adma.202510419

- Zhiying Liu, **Ganlin Chen**, Soumya S Dash, Yinghao Zhou, Changjun Cheng, Mingqiang Li, Jiahui Zhang, Huicong Chen, Weibing Wang, Weicheng Xiao, Daolun Chen, Ji-Jung Kai, Liang Qi, Yu Zou. Slip heterogeneity in a colony-structured titanium alloy: Planar versus wavy slip traces. ***International Journal of Plasticity***, 194, 104462 (2025) https://www.sciencedirect.com/science/article/pii/S0749641925002219
- **Ganlin Chen**, Dian Li, Yufeng Zheng, Liang Qi. Stability and Growth Kinetics of $\{112\}$ twin embryos in β-Ti alloys. ***Acta Materialia***, 263, 119520 (2024) https://doi.org/10.1016/j.actamat.2023.119520
- Aditya Sundar, **Ganlin Chen** and Liang Qi. Substitutional Adsorption of Chloride at Grain Boundary Sites of Hydroxylated Alumina Surfaces Accelerates the Initiation of Localized Corrosion, ***npj Materials Degradation*** 5, 1-15 (2021) https://doi.org/10.1038/s41529-021-00161-w
- Haoyue Wangchen, **Ganlin Chen**, Le Wei, Yumo Pan, Yinghui Sun and Liang Zhao. Research Progress for Graphene Based Anode Materials of Lithium ion Battery. ***Chinese Battery Industry*** 23, 269-275 (2019)

PRESENTATIONS

Oral Presentations

- The 10th International Conference on Multiscale Materials Modeling, Baltimore, Maryland, 2022
- The 3rd World Congress on High Entropy Alloys, Pittsburgh, Pennsylvania, 2023
- Materials Research Society Fall Meeting & Exhibit, Boston, Massachusetts, 2024
- 155th TMS Annual Meetings & Exhibition,San Diego, California, 2026

Poster Presentations

- Michigan Materials Research Institute Annual Summit, Ann Arbor, Michigan, 2026
- Physical Metallurgy Gordon Research Conference, Easton, Massachusetts, 2025
- Physical Metallurgy Gordon Research Seminar, Easton, Massachusetts, 2025
- Michigan Materials Research Institute Annual Summit, Ann Arbor, Michigan, 2024
- Summer School for Integrated Computational Materials Education, Ann Arbor, Michigan, 2024

TECHNICAL SKILLS

Programming Languages: Python, C++, Matlab
Libraries and Tools: PyTorch, Julia, Pandas, Numpy, Seaborn
Technical Software: Thermo-Calc, PIUMED, LAMMPS, VASP, ABAQUS

EXTRACURRICULAR ACTIVITIES

- **Voluntary Teacher** in Kenya, Africa *Aug. 2017*